## 7.9 Factors affecting the reaction rate:

The rate of a reaction is affected by the following factors.

1. Nature and state of the reactant
2. Concentration of the reactant
3. Surface area of the reactant
4. Temperature of the reaction
5. Presence of a catalyst

### 7.9.1 Nature and state of the reactant:

We know that a chemical reaction involves breaking of certain existing bonds of the reactant and forming new bonds which lead to the product. The net energy involved in this process is dependent on the nature of the reactant and hence the rates are different for different reactants.

Let us compare the following two reactions that you carried out in volumetric analysis.

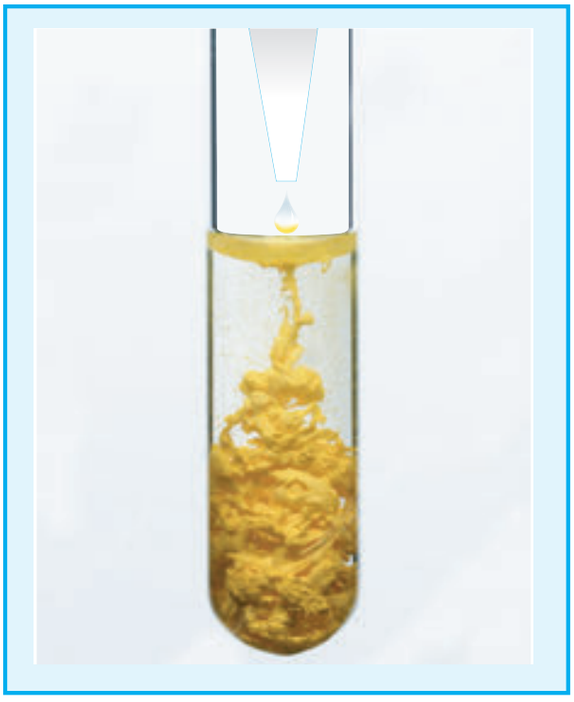
1). Redox reaction between ferrous ammonium sulphate (FAS) and $KMnO_4$

2). Redox reaction between oxalic acid and $KMnO_4$

The oxidation of oxalate ion by $KMnO_4$ is relatively slow compared to the reaction between $KMnO_4$ and $Fe^{2+}$. In fact heating is required for the reaction between $KMnO_4$ and Oxalate ion and is carried out at around $60^0C$.

The physical state of the reactant also plays an important role to influence the rate of reactions. Gas phase reactions are faster as compared to the reactions involving solid or liquid reactants. For example, reaction of sodium metal with iodine vapours is faster than the reaction between solid sodium and solid iodine.

Let us consider another example that you carried out in inorganic qualitative analysis of lead salts. If you mix the aqueous solution of colorless potassium iodide with the colorless solution of lead nitrate, precipitation of yellow lead iodide take place instantaneously, whereas if you mix the solid lead nitrate with solid potassium iodide, yellow coloration will appear slowly.



### 7.9.2 Concentration of the reactants:

The rate of a reaction increases with the increase in the concentration of the reactants. The effect of concentration is explained on the basis of collision theory of reaction rates. According to this theory, the rate of a reaction depends upon the number of collisions between the reacting molecules. Higher the concentration, greater is the possibility for collision and hence the rate.



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